The AI Dependency Index (IDA): Quantifying Minimum Workforce Continuity in AI-Dependent Organizations

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Abstract

Problem. Organizations adopting artificial intelligence compute how many people they can release. I found no published framework quantifying how many they must retain to keep operating if the provider becomes unavailable. On 12 June 2026 a United States export-control order required a frontier provider to suspend two of its most advanced models for all foreign nationals; unable to verify nationality per user, it disabled them for everyone until 30 June [6]. The cause was policy, not infrastructure. **Gap.** Resilience frameworks prescribe dependency mapping and fallbacks without sizing them. The AI Resilience Framework [1] leaves a quantitative substitutability score undeveloped; it is developed here in workforce terms. **Method.** This paper proposes the AI Dependency Index (IDA), a deterministic model on six operational inputs, a declared continuity floor, the AI expense and three project parameters. It returns a retained workforce, a risk-adjusted saving, a defensible investment ceiling and a 0–100 exposure score. **Findings.** Past the continuity floor, further automation raises the AI bill without releasing anyone: savings fall while exposure rises. Under fixed cost they plateau, yet exposure still rises. The best level of automation sits exactly where that floor starts to bind, $a^* = 1 - P_{min}/P$. **Robustness.** The fragility coefficients were set by judgement, not measurement. The calculation was repeated 20,000 times with each moved at random by up to 50 per cent either way. **That point did not move**, nor did the ordering between architectures. The verdict on an individual project did: the model therefore reports the threshold at which it flips, rather than a yes or a no. **Availability.** Model, formulas, implementation, 25 edge cases and robustness script are published under the MIT licence with a permanent identifier [2]. At **modelo-ida.web.app**.

Keywords: operational resilience; business continuity; AI dependency; vendor concentration; workforce planning; decision support systems.
JEL: M15, O33, L86, D81.

1. Introduction

Every business case for automation answers the same question: how many people will this let us do without. The question is old, the arithmetic is familiar, and the answer feeds directly into a return-on-investment calculation that boards know how to read.

There is a second question that the same business case does not ask. If the automated work now depends on an external provider, and that provider stops being available, how many people does the organization need in order to keep operating at all? This is not the first number with the sign reversed. It is a different quantity, governed by a different constraint, and in current practice it is simply not computed.

The asymmetry was tolerable while the risk was hypothetical. Two developments in 2026 make it concrete.

First, price. The amount an organization pays per unit of inference is not anchored to a verifiable cost structure that the buyer can inspect or predict. It is set unilaterally by providers whose own business models are not yet settled, which means the operating cost of an automated process can move for reasons entirely outside the adopter's control and unrelated to its own consumption.

Second, availability, and this one is no longer hypothetical. On 12 June 2026 the United States Department of Commerce issued an export-control directive to Anthropic requiring it to suspend access to two of its most advanced models, Claude Fable 5 and Claude Mythos 5, by any foreign national, whether inside or outside the United States [6]. Unable to verify the nationality of each user in real time, the provider disabled both models for every customer worldwide [7]. The controls were lifted on 30 June, nearly three weeks later, and access was restored in stages [8]. No infrastructure failed. No service-level agreement was breached in the ordinary sense. A multi-cloud infrastructure would not by itself have addressed the restriction, because the constraint was not technical. The provider's other models remained available, but that is not the same as a fallback:

switching models is not automatic. It requires monitoring that raises the alarm, an engineer to change the code where the model identifier is fixed in it, and a switch that has been rehearsed beforehand. Absent any of the three, the interruption occurs anyway and lasts as long as it takes people to resolve it — which is again measured in hours of human capacity. Section 6 returns to that distinction, which the present model does not yet price.

An organization that had automated a critical process on that provider had, for those three weeks, exactly as much operating capacity as it had retained people who could still do the work by hand. Very few of them knew that number in advance.

Provenance. The model condenses roughly a decade of consulting practice on AI and automation adoption across regulated and operationally critical sectors, during which its formulation was repeatedly revised against engagements that cannot be identified here for reasons of client confidentiality. That history is the reason the model asks for six operational inputs rather than sixty: every input that a manager could not answer from memory in a meeting was removed. It is also, necessarily, a limitation on the evidence this paper can offer: the practice informed the design but is not reported here as data, and the published case in Section 4 is constructed rather than observed. Independent empirical validation therefore remains open, and Section 6 states what would close it.

Contribution. This paper proposes the AI Dependency Index (IDA), a deterministic model that computes the minimum workforce an organization must retain for continuity, the saving that remains once that constraint and architectural fragility are priced in, the maximum investment such a project can justify, and a bounded index of operational exposure. The model is deliberately small: its operational core is six inputs that any operations manager already possesses, requires no dataset and no training, and is published in full so that its results can be reproduced or refuted.

2. Related work and the gap

Three bodies of work bear on the problem, and none of them returns a headcount.

Governance standards. ISO/IEC 42001 [3] establishes requirements for an AI management system, including risk assessment and treatment. The European Digital Operational Resilience Act (DORA) [5] obliges financial entities to map information and communication technology dependencies and to test resilience against severe but plausible scenarios. Both are prescriptive about process and silent about magnitude: they require that a contingency plan exist, not that it be sized.

Regulatory evidence on concentration. The dependency this paper sizes is already being measured by supervisors. In the 2024 Bank of England and Financial Conduct Authority survey of UK financial services, a third of all AI use cases were third-party implementations, against 17 per cent two years earlier, and the top three third-party providers accounted for 73, 44 and 33 per cent of all reported cloud, model and data providers respectively [9]. The direction of travel matters more than the levels: the share held by the top three cloud providers fell relative to 2022, while the share held by the top three model providers rose from 18 to 44 per cent [9]. Concentration is migrating from the infrastructure layer to the model layer. The Financial Policy Committee has since set out, as a financial-stability scenario, that where customer-facing functions have become heavily reliant on vendor-provided AI models, a widespread outage of one or several key models could leave many firms unable to deliver vital services such as time-critical payments [10]. That scenario was published in April 2025, fourteen months before the episode described in Section 1.

The policy response has been institutional rather than quantitative. The Financial Services and Markets Act 2023 established a regime for critical third parties, in response to the Committee's 2021 finding that concentration in the provision of services to UK firms could raise financial-stability risk without additional measures; the Bank, the Prudential Regulation Authority and the Financial Conduct Authority published joint rules for that regime in November 2024 [10][11]. In January 2026 the House of Commons Treasury Committee recommended that HM Treasury designate the major AI and cloud providers as critical third parties by the end of 2026 [12]. None of these instruments states how many people a firm must retain in order

to operate through the scenario the Committee describes. That is the quantity this paper computes.

Business continuity practice. Established continuity methodology already provides the conceptual apparatus for a minimum operating level, most directly through business impact analysis and the notion of a minimum business continuity objective [4]. What it does not provide is a way to derive that level from the specific structure of an AI dependency, where the substituting resource is human labour that the automation decision itself is in the process of removing.

Recent AI-resilience literature. The closest published work is the AI Resilience Framework [1], which argues that governance centred on trustworthiness leaves an obligation of resilience inadequately covered, and proposes dependency mapping, tiering by criticality and substitutability, fallback protocols, and management of provider concentration. That framework explicitly identifies a quantitative substitutability score as a natural extension that would allow tiering thresholds to be calibrated rather than asserted, and does not develop it. The present work takes up that extension and expresses substitutability in the only unit in which a fallback protocol can actually be executed: people who can do the work.

Adjacent index work exists but answers a different question. Firm- and industry-level indices of AI transformation measure opportunity, disruption risk and value creation [13]; they describe how much an organization stands to gain, not how much operating capacity it would retain if the technology became unavailable.

One recent result does reach an over-automation conclusion, by an entirely different route. Falk and Tsoukalas [14] show, in a competitive task-based model, that a demand externality drives rational firms to automate beyond the collectively optimal level, because each firm captures the full saving from replacing a worker while bearing only part of the resulting loss of demand. The present model reaches an over-automation result at a different scale and without requiring an externality: past the continuity floor a single firm automates beyond its own optimum, for reasons internal to its own accounts. The two are complementary, one macroeconomic and one operational, and both contradict the premise that maximum automation is optimal automation.

Across those three bodies of work I did not find a published framework that returns a headcount. The prescription is qualitative, while the decision a manager has to sign is quantitative.

3. The model

One case runs through this section so that every quantity arrives with a number attached. **An organization automates the handling of customer enquiries.** The area has ten people and a monthly payroll of 16,000, each working 160 hours a month. Sixty per cent of the work could be done by the system, a fifth of that requires sign-off by a qualified professional, and in production the system performs at eighty per cent. Six hundred and forty hours a month have to be covered whatever happens. Section 4.1 returns to the same case with every figure in one table. No sector is named because none is assumed: the model asks the same questions of a bank, a hospital or a logistics operator.

Two layers of this model are worth separating at the outset, because they do not rest on the same footing. The first is **structural**: a_{ef}, P_{min}, P' and the threshold a^* of Section 4.2 follow from counting hours and dividing them by hours. They contain no coefficient chosen by the author, and their two conclusions — that a floor exists, and that beyond a certain level further automation releases nobody — hold for anyone who accepts the definitions. The second is **heuristic**: λ, S and the index incorporate premiums set by expert judgement, and their absolute values are exactly as debatable as those premiums.

The distinction governs how the rest should be read. A reader who rejects Table 2 outright may discard the second layer and keep the first intact. Section 4.5 measures how much the second layer withstands; the first needs no such defence.

3.1 Inputs

The model takes six operational quantities, listed first because they are the ones the design was constrained by: each must be answerable by an operations manager without a data project. Five further quantities are required, of a different kind, and a sixth is optional. H_{min} is a declaration of policy rather than a measurement; C is read off an invoice; I, M and T_{max} describe the project rather than the operation. The architecture premiums of Section 3.5 are selected from a closed list, not entered as numbers. The optional term is the external backup R of Section 3.6, which is admitted only under the three conditions stated there and is the only input that lowers reported exposure. Table 1 gives all of them.

Symbol	Input	Unit
P	Headcount of the area under analysis	people
N	Monthly payroll of that area	currency/month
h_p	Working hours per person per month	hours/month
a	Share of the work that AI could perform	fraction
r	Hours of that work a person retains regardless ($r_n + r_p$)	fraction
η	Effectiveness actually achieved by the AI	fraction
H_{min}	**Hours per month that must be covered whatever happens**	**hours/month**
C	**Monthly AI expense**	**currency/month**
I	**Initial investment**	**currency**
M	**Monthly maintenance**	**currency/month**
T_{max}	**Horizon at which the organization approves investments**	**months**
R	**Verified external backup (Section 3.6)**	**people**

Table 1. All model inputs. The first six, in roman, are the operational ones the design was constrained by. The remainder, in bold, are a policy declaration (H_{min}), an invoice figure (C), three project parameters and the optional backup term. The architecture premiums of Table 2 are selected from a closed list rather than entered as numbers. Earlier presentations of this model described it as taking six inputs; that count refers to the operational block alone.

3.2 Effective automation

The share of work that is genuinely automatable is smaller than the share that is technically automatable. Two discounts apply, and they are independent of each other.

The first, **r**, is what is not delegated. It may be required by regulation — a law or a supervisor demands that a qualified person authorise it — or decided by the organization itself: the system triages the complaint, but a person answers when the customer threatens legal action. Both discount identically, though one comes from outside and the other from within.

Both origins discount identically, but they are worth writing separately because they do not rest on the same footing: $r = r_n + r_p$. The first term is **exogenous** — the sector fixes it, the organization did not choose it and cannot remove it. The second the organization set, and the organization can undo. A telephone contact centre in healthcare or in a police force has hours a person retains because the regulation says so; the same centre, handling the same enquiries, in a retail chain may have $r_n = 0$ because there is nothing to authorise. **r is not a property of the task but of the sector in which the task takes place.**

One consequence is worth stating explicitly, because it is usually read the other way round: regulation operates as continuity insurance. The same automation project, identical in every other respect, yields a lower index in the hospital than in the retail chain, because the regulation retains human hours that nobody may delegate. What is felt in the short run as a constraint is, in this model's accounting, exposure that was never taken on.

The policy term enjoys no such stability. It was a decision, and decisions get revisited — characteristically under cost pressure, which is exactly when those human hours are needed. Two organizations with the same index are therefore not equally solid: the one whose r is entirely policy rests its position on something a board can reverse in a single meeting. The reference implementation asks for the split as an **optional reporting input**: it enters no calculation, because the two terms sum, and it is reported alongside the index so that the difference in durability is visible.

A point that is frequently confused: **r does not depend on the quality of the system**. Even were the model to perform perfectly, those hours would remain human. r and η are independent by construction — improving the system raises η and does not move r by a single point — and so no technical advance, however large, eliminates the hours that r retains.

r is measured in **hours retained**, not in tasks a person touches. The distinction changes the result. If the system drafts the reply in thirty minutes and a person authorises it in thirty seconds, that task contributes the thirty seconds and not the thirty minutes: what was retained is the authorisation, not the work. Counting the whole task inflates r, depresses a_{ef} and inflates P′ — that is, it retains people the model is not asking for. This is the likeliest measurement error in practice, because the list of tasks requiring human intervention is easy to obtain and the distribution of hours within each task is not.

The second, **η**, is how well the system performs in production rather than in demonstration. It is not the model's accuracy: it is the fraction of the delegated hours that stop being human work. The two numbers diverge more than they appear to. A system correct on eight cases out of ten may deliver well under 0.8, because finding the two that are wrong means reviewing all ten, and that review is human work that was not eliminated. η should therefore be estimated from hours actually saved, not from a reported accuracy rate. The two are distinct: r bounds *what* may be delegated, η bounds *how well* what was delegated gets done.

$$a_{ef} = a \cdot (1 - r) \cdot \eta$$

The composition is multiplicative, and the resulting gap is not a rounding error. The case of this section believes it is automating 60 per cent of the process. A fifth of those hours is retained regardless, and the system performs at 80 per cent, so the share actually automated is 0.60 × 0.80 × 0.80 = **38.4 per cent** — roughly two thirds of what was assumed. Every subsequent quantity is computed on a_{ef}, never on a.

3.3 The continuity floor

The organization declares H_{min}: **the hours of work per month that have to be covered even with no AI at all.** If those hours are not covered, the process does not happen. This is a statement of policy or regulation, not an estimate the model derives.

There is a third source, and in practice it is the most common: the **contract**. An organization that signed a service level agreement committed to answering, and that commitment says nothing about which tool. The customer did not accept "we will respond unless our AI provider is unavailable": it accepted a response time. H_{min} then captures whatever must be covered to sustain what was signed, and the question that organizes the estimate is direct: **if the system is gone tomorrow, how much of what we committed to remains enforceable?** The answer is almost never "none".

It is answered by counting, not by modelling. The organization of this section reasons as follows: if the system stops, 800 enquiries a month still have to be answered and one person answers two an hour, which is 400 hours; someone also has to review orders eight hours a day, which is 240 more. H_{min} = **640 hours**. The floor in people follows directly.

$$P_{min} = ceil(H_{min} / h_p)$$

3.4 Retained workforce

Economics alone would set headcount by the share of work still done by people; call that P_{econ}. Continuity sets a lower bound that economics may not cross, which is P_{min}. **The retained workforce, written P′, is the**

larger of the two — and P′ is the number this paper exists to compute.

$$P_{econ} = ceil(\, P \cdot (1 - a_{ef}) \,) \qquad P' = max(\, P_{econ} \,,\, P_{min} \,)$$

The maximum operator is where the model departs from a conventional business case, and it is the source of every non-obvious result in Section 4. When P_{min} binds, additional automation stops converting into released people while continuing to convert into cost.

A boundary case deserves explicit treatment. If $P_{min} > P$, the organization is already below the headcount required to operate without the AI. The model reports this rather than clamping it: it is a finding about the present state, not an error in the inputs.

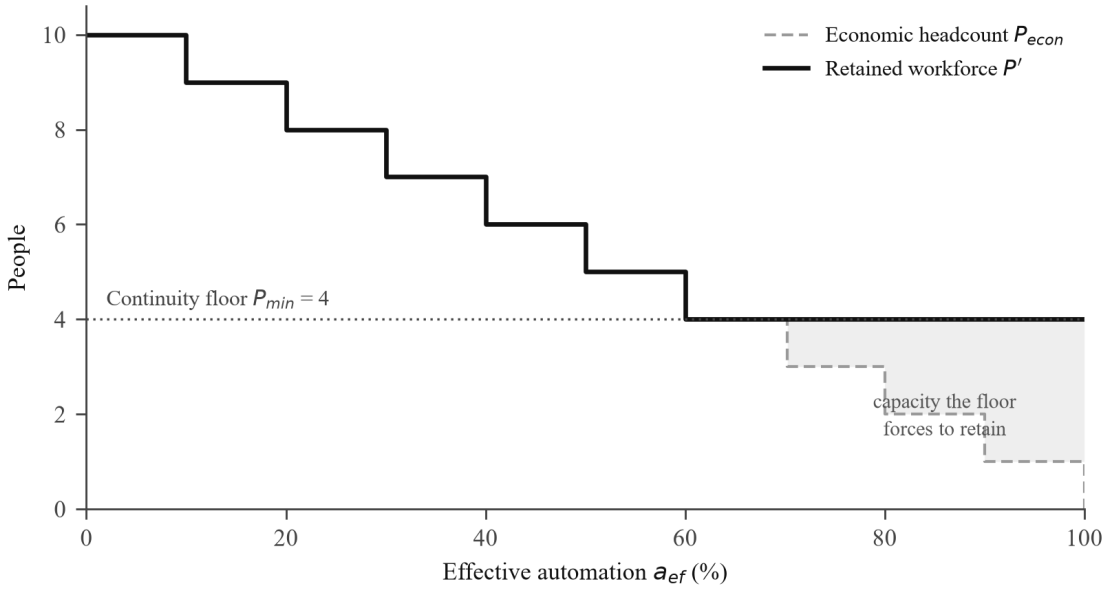


Figure 1. The continuity floor. Economic reasoning alone would keep reducing headcount as effective automation rises (dashed). The retained workforce (solid) follows it only until it reaches the floor the organization declared, and then stops. The shaded area is the labour capacity that continuity forces the organization to keep, and that a conventional business case would have released. Reference configuration of Section 4.2.

3.5 Risk multiplier and risk-adjusted saving

Architectural fragility is priced as a surcharge on the AI expense, written λ. The multiplier starts at unity and accumulates one premium per declared fragility.

The surcharge is not an invoice. It is the logic of motor insurance: driving uninsured is not cheaper, it costs the same until the crash. The cost of a fragile architecture arrives in full on a single day, and λ spreads it into instalments so that the architecture looks expensive while it is still cheap to change.

$$\lambda = 1 + \Sigma \rho_i$$

Condition	Premium ρ_i
AI runs on the organization’s own infrastructure	0.02
AI runs across multiple cloud providers	0.10
AI runs on a single cloud provider	0.30
No documented contingency plan	+0.20
Sensitive data processed outside the organization	+0.15

Table 2. Architecture premiums. The first three are mutually exclusive; the last two are additive. The attainable maximum is λ = 1.65. These coefficients are declared expert judgement; see Section 6.

The AI expense behaves in one of two ways, and the distinction governs Section 4. Under usage-based billing it scales with automated hours, $C = c \cdot P \cdot h_p \cdot a_{ef}$, where c is the expense per automated hour; under fixed-cost architectures C is a constant. Writing $C_h = N / (P \cdot h_p)$ for the cost of a human hour, the model is meaningful only when $\lambda \cdot c < C_h$, that is, when the risk-adjusted AI hour is cheaper than the human hour it replaces; otherwise the organization is paying to automate. In the reference case of Section 4.1, c = 2 and $C_h = 10$.

Two qualifications about that condition and about c. First, $\lambda \cdot c < C_h$ is **necessary but not sufficient** for the saving to be positive. People are released one whole person at a time while the AI expense grows continuously, so between two steps the organization pays for automation that has not released anyone yet. In the reference case it pays for 3.84 person-equivalents of AI and releases 3, so the break-even on $\lambda \cdot c$ falls from 10 to **7.81** before maintenance is counted. Where $P \cdot a_{ef} < 1$ the saving is outright negative, because nobody is released and the invoice already runs.

Second, c is defined **per human hour actually saved**, not per hour delegated to the system. The distinction matters when estimating it. A vendor charges for what it is sent, which is the delegated hours $a \cdot (1 - r)$; the hours that stop being human are fewer, because η discounts the review. Taking the vendor's rate per delegated hour and using it as c understates the invoice by a factor of η. Anyone preferring to work from the list rate should therefore divide it by η, and note that doing so makes c and η dependent — which the simulation of Section 4.4 assumes away when it samples them separately.

The risk-adjusted saving subtracts the surcharged AI cost and maintenance from the payroll of the people actually released.

$$S = (P - P') \cdot (N / P) - C \cdot \lambda - M$$

S is an economic evaluation quantity, not a cash-flow projection. The organization does not write a cheque for $C \cdot \lambda$; the surcharge exists so that a fragile architecture appears expensive while it is still cheap to change. An accountant will compute a larger saving, namely the same expression with $\lambda = 1$, and both numbers are correct for their respective purposes.

3.6 Payback, investment ceiling and external backup

Simple payback follows as $T = I / S$. Inverting it against the horizon at which the organization approves investments yields the quantity that has proved most useful in practice.

$$I_{max} = S \cdot T_{max}$$

Both quantities are defined only for $S > 0$. Where the risk-adjusted saving is zero or negative the project has no finite payback and no positive investment ceiling, and the model reports that rather than a negative figure; the Monte Carlo of Section 4.4 counts those runs separately for the same reason.

This reframes the decision. Rather than reporting that a project pays back in twenty months and inviting a binary approval, the model states the maximum amount the project can justify, which is a negotiating position computed from the organization's own saving and its own risk.

Sensitivity of the ceiling. Because S carries the surcharge, I_{max} inherits the expert-judgement coefficients of Table 2 and should be read as a band rather than a point. In the reference case of Section 4.1 the ceiling ranges from 44,222 at the declared fragility to 55,282 with no surcharge at all.

Architecture	λ	S	I_{max}	Project of 50,000
Single cloud, no contingency plan	1.50	2,456.8	44,222	rejected
Single cloud, with plan	1.30	2,702.6	48,646	rejected
Multi-cloud, with plan	1.10	2,948.3	53,070	approved
Own infrastructure, with plan	1.02	3,046.6	54,839	approved
$\lambda = 1$ — what an accountant would compute	1.00	3,071.2	55,282	approved

Table 3. The investment ceiling as a function of architectural fragility, holding the AI invoice constant at 1,228.80 per month so that λ is isolated. Under an on-premises deployment the invoice itself would change; the row isolates the multiplier, not the architecture as a whole.

Solving $I_{max} = I$ for λ gives the point at which the verdict changes. Here it is λ = 1.24.

That number is more useful than an approval. The model does not say yes or no. It says: **this project is worth its price if, and only if, running on a single provider without a rehearsed fallback costs less than an extra 24 per cent on the AI bill.** That is the question to put to a board, because it is the one question the model cannot answer on the board's behalf. Reporting the threshold instead of the verdict makes explicit that the uncalibrated coefficient is what is being endorsed. Section 4.5 establishes which conclusions do not depend on that endorsement.

Exposure is discounted by verified external backup. R is the number of people outside the analysed area who can perform the work on the day of an outage, without training.

$$coverage = min(R / P_{min} , 1) \quad (coverage = 0 \text{ when } P_{min} = 0)$$

Two design decisions constrain R. First, it is admitted only when three conditions are jointly declared: that those people have performed this process, that the manual procedure is documented, and that the handover has been rehearsed. If any is absent, R is forced to zero. R is the only input that lowers the reported risk, and is therefore the one most exposed to optimistic reporting. Second, coverage discounts exposure but never eliminates it: the maximum discount is 0.7, because people brought in from elsewhere abandon their own posts, do not perform the task daily, and lack accumulated context.

It should be noted that R does not reduce P′. External backup lowers how exposed the organization is, not how many people it needs. This is deliberate: **R is treated as an emergency mitigation, not as an ex ante justification for reducing the area's permanent workforce**.

3.7 The index

The index composes the three factors that determine exposure: how much of the work now depends on the system, how fragile the architecture carrying it is, and how much of the continuity floor is externally covered.

$$IDA = min(a_{ef} \cdot \lambda \cdot (1 - coverage \cdot 0.7) \cdot 100 , 100)$$

The bounds are interpretable: 0 denotes no measurable dependency, 100 denotes the maximum dependency the model represents. The cap is required because the unbounded product reaches 165 at $a_{ef} = 1$ with λ = 1.65; everything above 100 is already maximal exposure, and truncating preserves a stable meaning for the scale. Reported bands are 0–30 resilient, 30–60 watch, 60–100 critical.

Truncation carries a cost, and it is worth declaring. Two organizations whose unbounded product is 105 and 165 both report an index of 100, and yet the second is considerably more exposed than the first. Where that difference matters — comparing business units against each other, or tracking one organization over time — the **gross exposure** $E = a_{ef} \cdot \lambda \cdot (1 - \text{coverage} \cdot 0.7) \cdot 100$, without the minimum, should be reported alongside the index. The index remains the figure that is communicated; E is the one that preserves the ordering above the cap.

Saving and exposure are reported separately and are not combined into a single score, because they move independently. Two organizations may save the same amount every month and be in entirely different positions: one has automated a third of its work across two providers and scores in the low twenties; the other has automated nearly everything on a single provider with no rehearsed fallback and scores near a hundred. The monthly saving does not distinguish them. Collapsing the two numbers into one would conceal precisely the trade-off the model exists to expose.

4. Properties and worked example

4.1 Reference case

The case of Section 3, in full. An organization automates the handling of customer enquiries. The area has ten people, a monthly payroll of 16,000, and 160 working hours per person. Sixty per cent of the work is automatable, of which twenty per cent requires sign-off by a qualified professional; the system performs at eighty per cent. Six hundred and forty hours per month must be covered whatever happens. The project costs 50,000 with 500 per month of maintenance, runs on a single cloud provider with no contingency plan, and is evaluated against an eighteen-month horizon.

Quantity	Value
Effective automation a_{ef}	0.384
Continuity floor P_{min}	4 people
Economic headcount P_{econ}	7 people
Retained workforce P'	7 people (3 released)
Risk multiplier λ	1.50
Risk-adjusted saving S	2,456.80 per month
Payback T	20.4 months
Investment ceiling I_{max}	44,222
Index IDA	58 (watch)

Table 4. Reference case. The decisive line is the last but one: the project was presented at 50,000 and the model states that it is worth at most 44,222 to this organization at this declared level of risk. Table 3 gives the band around that figure, and the threshold at which the verdict reverses.

4.2 An automation optimum exists under variable cost

Setting $r = 0$ and $\eta = 1$ to isolate the effect, and letting the automated share rise, produces a result that contradicts the intuition the business case is built on.

Effective automation	Naive promise	Actual saving	Payback	IDA
60% (optimum here)	6,220	**6,220**	**8 months**	90
80%	8,460	5,260	10 months	100
100%	10,700	4,300	12 months	100

Table 5. Past the continuity floor the organization becomes poorer and more dependent simultaneously. The naive column assumes every automated hour is a salary removed.

The mechanism is the maximum operator of Section 3.4. Once P_{econ} falls below P_{min}, the floor binds: released headcount stops growing while automated hours, and therefore the AI bill, continue to grow. Savings decline. The location of the optimum is not case-specific; it admits a closed form.

Proposition (the level at which the continuity floor first binds). For $0 \le P_{min} \le P$, under usage-based billing with $\lambda \cdot c < C_h$, and with the headcount and cost structure of Section 3, the risk-adjusted saving S is maximised at

$$a^* = 1 - P_{min} / P$$

the smallest effective automation at which the continuity floor binds, and S is strictly decreasing for $a_{ef} > a^*$. The saving at the optimum is

$$S(a^*) = (P - P_{min}) \cdot h_p \cdot (C_h - \lambda \cdot c) - M$$

Proof sketch. Released headcount $k = P - P'$ is a step function of a_{ef}: k people are released for a_{ef} in [k/P, (k+1)/P), up to the maximum $k = P - P_{min}$ imposed by the floor. Within each step the payroll term is constant

while the cost term increases linearly, so S attains its maximum on the step at its left edge, where $S = k \cdot h_p \cdot (C_h - \lambda \cdot c) - M$. This is increasing in k whenever $\lambda \cdot c < C_h$, so the global maximum is at the largest attainable k, whose left edge is $a^* = (P - P_{min})/P$. Beyond a* no further headcount is released while cost continues to accrue. □

Two qualifications bound the proposition. The first is its domain: when $P_{min} > P$ the expression returns a negative value and no longer denotes a level of automation, because the organization is already below the workforce it requires — the edge case of Section 3.4. The second is feasibility: a* is the **structural** optimum, while attainable effective automation has a ceiling of its own, $a_{ef}^{max} = a \cdot (1 - r) \cdot \eta$ at a = 1. If that ceiling lies below a*, the floor never binds : with r = 0.20 and η = 0.80 the ceiling is 0.64, only just above the case's a* = 0.60.

Where the maximum then sits requires care, because the saving is a step function. Within each step the released headcount does not change while the AI cost does, so S decreases: the attainable maximum sits not at the ceiling but at the **left edge of the highest step the ceiling allows**, at $\text{floor}(P \cdot a_{ef}^{max}) / P$. With r = 0.30 and η = 0.80 the ceiling is 0.56 and that edge is 0.50: operating at 0.50 releases five people and leaves **5,100** a month, while operating at 0.56 releases the same five and leaves **4,812**. The 288 difference is AI paid for without releasing anyone. The two points coincide only when $P \cdot a_{ef}^{max}$ is an integer. The **structural** and **feasible** optima therefore coincide when $a^* \leq a_{ef}^{max}$.

The point is defined by the floor rather than by an optimality argument: it is the smallest effective automation at which P_{min} begins to bind, and under the stated assumptions it coincides with the maximum of risk-adjusted saving. In the reference configuration $a^* = 1 - 4/10 = 0.60$ and $S(a^*) = 6 \cdot 160 \cdot (10 - 3) - 500 = 6{,}220$, which is the first row of Table 5. **The location of the optimum therefore depends on the declared floor and on headcount alone.** Wages, unit cost and λ change its height, not its position — a fact Section 4.5 puts to work. Under fixed cost the same a* marks the start of the plateau of Section 4.3, and since $I_{max} = S \cdot T_{max}$, the investment ceiling is maximised at the same point. Table 5 sets r = 0 and η = 1 to make a_{ef} coincide with a; the proposition itself is stated on a_{ef} and holds for any r and η. This is a firm-level counterpart to the over-automation result of [14].

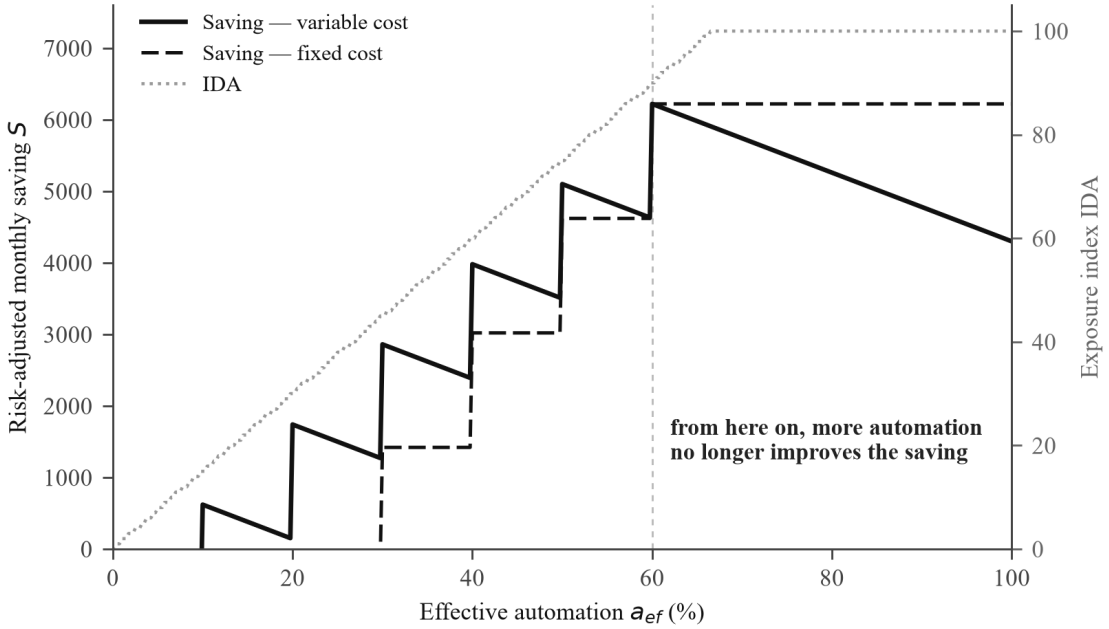


Figure 2. Saving against exposure. Once the floor binds at 60 per cent, the two cost structures diverge: under variable cost the saving declines, because automated hours keep accruing while released headcount does not; under fixed cost it plateaus. The exposure index rises in both cases. To the right of the dashed line, further automation buys no additional saving in either regime and continues to buy dependency. The fixed cost is set to the variable-branch expense at $a_{ef} = 0.60$, so that both curves coincide there and the comparison isolates the cost structure rather than the level of spend.

4.3 Fixed-cost architectures plateau rather than decline

The preceding result assumes the AI is billed per unit of use. Where the organization operates its own infrastructure, C is a constant, and since released headcount is non-decreasing in a_{ef} while every other term is fixed, S is non-decreasing. In the same configuration as Table 5 with a fixed monthly cost, the saving is identical at 60 and at 100 per cent effective automation.

The consequence is not that the trade-off disappears but that it changes currency. Under fixed cost, automating past the floor no longer makes the organization poorer, yet the index still rises from 90 to 100 across the same interval. Additional automation purchases dependency for nothing. Under variable cost the organization loses money and resilience together; under fixed cost it loses only resilience, which is harder to see in a set of accounts and more expensive to repair afterwards.

The distinction this makes visible is worth stating plainly: **automation does not imply autonomy**. An organization can automate an increasing share of its work and, at the same time, become less able to operate on its own. The two quantities move independently, and only one of them is usually measured.

4.4 Uncertainty

Point estimates of a, η, unit cost and λ are rarely defensible at the moment a project is approved. A Monte Carlo layer draws each of them independently from a uniform distribution over a range the analyst declares, holds the remaining inputs at their declared values, and reports the proportion of runs in which the investment is recovered within T_{max} under the declared scenario. The retained share r is deliberately not sampled: it is a decision already taken, and changing it is a new scenario to be simulated rather than a random draw. Neither are P, N, h_p and H_{min} sampled: they are quantities the organization holds or declares, not uncertainties about it. Drawing them at random would be simulating a different organization rather than the uncertainty facing this one.

Sampled quantity	Distribution	Range
Automatable share a	uniform	0.50 – 0.70
Effectiveness η	uniform	0.65 – 0.90
Cost per automated hour c	uniform	1.50 – 4.00
Risk multiplier λ	uniform	1.30 – 1.70

Table 6. Sampling specification for the reference case. Held fixed at the values of Section 4.1: r = 0.20, P = 10, h_p = 160, N = 16,000, P_{min} = 4, M = 500, I = 50,000, T_{max} = 18. Draws: 10,000, with the mulberry32 generator and seed 7. The λ range extends slightly beyond the 1.65 attainable from Table 2 because the simulation treats fragility as a continuous uncertainty rather than a selection from the closed list of premiums.

Over 10,000 draws the investment is recovered within the eighteen-month horizon in **25.4 per cent** of runs, with 0.4 per cent returning a negative saving and 6.4 per cent recovering only beyond sixty months. That figure, not its precision, is the finding: at these ranges the project is closer to a wager than to a plan, and the lever that moves the number is rarely more automation but lower architectural fragility or a smaller initial investment.

One feature of those ranges is worth stating, because it changes how everything above should be read. With a between 0.50 and 0.70 and η between 0.65 and 0.90, effective automation is bounded between 0.26 and **0.504**, always below this case's $a^* = 0.60$. **The continuity floor binds in none of the 10,000 draws.** The simulation therefore measures the project's uncertainty in the regime before the floor, and does not put the result of Section 4.2 to the test. That raising a or η rarely improves payback within these ranges is due not to the floor but to the $(1 - r)$ discount and to the step: over much of the sampling box the released headcount is stuck at three while the AI expense keeps climbing. A scenario that does exercise the floor calls for appreciably higher ranges of a and η, and is left to anyone repeating the exercise with the source.

Two qualifications govern how that number may be read. The ranges are **declared scenarios, not empirical probability distributions**: the uniform is a statement that the analyst regards the interval as plausible and holds no view within it, not a fitted density. The figure is therefore the share of runs that recover under the declared scenario, and not the probability that this project succeeds. Second, the four quantities are drawn independently, which is unlikely to hold: architecture and unit cost move together, and an organization that accepts a fragile architecture is often the one paying the least per hour. Correlated sampling may materially change the resulting distribution, and establishing how would require an empirical or scenario-based correlation structure that this work does not have. The independence assumption is an assumption, not a finding.

The implementation uses a seeded generator, so the figure is exactly reproducible: any reader opening the published implementation with these ranges obtains 25.4 per cent. The sampling error at this number of draws is 0.44 percentage points, and re-running under other seeds returns values between 24.6 and 25.9 per cent, consistent with that error. Reporting the seed matters here for a concrete reason: an earlier unseeded implementation of the same model returned 26 per cent, because the underlying value sits close to a rounding boundary. A Monte Carlo result quoted in a paper without its seed cannot be checked, only re-run.

4.5 Robustness of the conclusions to the premiums

The premiums of Table 2 are declared expert judgement, which invites the objection that the model's conclusions rest on numbers that were asserted. The objection cannot be answered by calibration here, but it can be bounded. Each premium was perturbed by ±50 per cent, both at the $3^5 = 243$ combinations of a three-level grid —each premium at −50 per cent, nominal or +50 per cent— and over 20,000 draws with each premium uniform on that interval, and four properties were examined.

Property	Result under ±50% perturbation
Location of the optimum a*	invariant — a single location for all λ
Existence of an interior optimum	holds while $\lambda < C_h/c = 5$: 3.0× the nominal maximum of 1.65, and 2.5× the 1.975 reached under a +50% perturbation
Ranking of the three mutually exclusive infrastructures, at fixed AI expense	preserved in 100.0% of 20,000 draws, but ±50% is the exact threshold at which inversion first becomes possible
Full ranking of all 12 configurations	identical in only 2.9% of the 243 combinations and 7.1% of draws

Table 7. Ordinal robustness. The first three properties survive; the fourth does not, and the reason is stated below.

The first property is immediate from the proposition of Section 4.2: $a^* = 1 - P_{min}/P$ contains no risk coefficient, and a numerical sweep of a_{ef} over [0,1] returns the same location for every attainable λ. The second follows from the condition $\lambda \cdot c < C_h$ of Section 3.5, which in the reference case is satisfied with a factor of three in hand. The third is the property the model is actually used for: choosing between architectures. It should be read as what it is, a ranking induced by the premiums of Table 2 while the AI expense is held constant, and not as a ranking of architectures in the round: an on-premises deployment changes the invoice itself, as the note to Table 3 records. What is shown to be stable is the ordering the declared premiums impose, not the full economics of switching provider.

The third result requires a qualification that strengthens rather than weakens it. Preserving the ranking in every draw is not evidence of a wide margin: two premiums $\rho_i > \rho_j$ can invert only when the perturbation factor f exceeds $(\rho_i - \rho_j)/(\rho_i + \rho_j)$, which for single cloud against multi-cloud is exactly 0.50. At ±50 per cent the architecture ordering is therefore at its boundary, not comfortably inside it, and the 100 per cent figure should be read as a tight bound rather than as slack. Inversion becomes materially likely only well beyond: the ordering survives in 97.7 per cent of draws at ±60 per cent and 73.9 per cent at ±100 per cent. The correct statement is that the premium-induced ordering is robust to any misestimation of the premiums smaller than a factor of three between single cloud and multi-cloud, that factor being their nominal ratio.

The exercise also surfaces an assertion embedded in Table 2 that is easier to see than to defend: the premium for the absence of a contingency plan, 0.20, is exactly the difference between single-cloud and multi-cloud provision. The model therefore claims that a rehearsed fallback is worth precisely as much as a second provider. That is a coefficient choice, not a finding, and it is among the first quantities the calibration exercise of Section 6 should test.

The fourth result is negative and is reported rather than omitted. The complete ordering of all twelve selectable configurations is not stable, but the instability is local and its cause is structural rather than statistical. Two pairs carry identical λ at the nominal values: multi-cloud without a plan ties with a single cloud with a plan at 1.30, and the same two with sensitive data tie at 1.45. Because the two pairs differ by the same quantity, the data premium cancels and **both ties always break in lockstep**, in the direction set by the sign of a single expression. That alone leaves the full ordering intact in half the draws, with nothing substantive having moved. The fall from 50 per cent to 7.1 is produced by other pairs, those that sit close without being tied: 0.03 separates 1.22 from 1.25, and three further gaps are 0.05. The median disturbance is four inversions out of sixty-six pairs. **The consequence for practice is that the model should not be used to separate two architectures whose multipliers differ by less than approximately 0.05.**

Taken with Section 3.6, the position is this: the structural conclusions of the model — where the optimum lies, which architecture is preferable — are robust to the coefficients, whereas verdicts on an individual project are not, and are for that reason reported as a threshold. The robustness analysis does not validate the premium values; it evaluates whether the model's qualitative conclusions depend critically on them. The premiums still require calibration; what this section establishes is which claims are waiting on it and which are not. The analysis is deterministic under a fixed seed — mulberry32 generator, seed 7, the same one the reference implementation uses — and distributed with the source.

One boundary of this exercise should be stated plainly, because it is easy to read the section as establishing more than it does. What is perturbed here are the *values* of the premiums, within the functional form the model already assumes: exposure as the product $a_{ef} \cdot \lambda$, discounted multiplicatively by coverage, and the surcharge applied to the AI expense. **Parameter uncertainty is not model-form uncertainty.** Nothing in this analysis establishes that the multiplicative composition is the correct one, that a weighted sum would not behave differently, or that exposure is the right quantity for those factors to compose into. Those are questions about the specification rather than about its coefficients, and the calibration exercise of Section 6 does not settle them either. They are the more interesting objection to this work, and this paper does not answer it.

5. Verification and reproducibility

The model is published together with an executable implementation that runs entirely in the browser, with no server component and no transmission of the data entered. Any reader can therefore reproduce every figure in this paper without installing software, registering, or exposing operational data to a third party. The implementation is available at **https://modelo-ida.web.app** and its source at **github.com/All-Aideas/Modelo-IDA**.

Correctness is checked by a battery of 25 edge cases distributed with the source. The battery does not compare the engine against itself: its expected values are computed by an independent re-implementation of the formulas exactly as published, so that any divergence between the documented model and the running code fails the test. The cases cover the boundaries where models of this kind break, including a floor exceeding current headcount, a single-person area, a thousand-person area, division by zero in the coverage term, an AI hour costing more than a human hour, both extremes of λ and of the approval horizon, declared but unverified backup, and both cost structures of Section 4.3. As of the date of this version the battery passes in full: 25 cases, no failures. The reference configuration returns $P' = 7$, $S = 2{,}456.80$, a payback of 20.4 months and an index of 58, and every figure reported in Section 4 is reproduced by the published implementation rather than transcribed into this document.

The implementation, the formulas, the test battery and this document are released under the MIT licence and deposited with a permanent identifier [2]. The robustness analysis of Section 4.5 is distributed with them as a standalone script with a fixed seed, together with its results.

6. Limitations

Nine limitations bear on how the results should be read, and are stated here rather than left for the reader to discover.

The coefficients are expert judgement. The premiums of Table 2 and the 0.7 discount cap were set from practice, not calibrated against a dataset of incidents. They are design parameters of the model. The immediate consequence is that comparisons within a single architecture are meaningful while absolute values across very different organizations should be treated with caution. Calibration against a corpus of documented provider outages is the most valuable extension available and the one I regard as the principal item of future work. Section 4.5 bounds, but does not remove, the consequences of this limitation. The reported bands — 0–30 resilient, 30–60 watch, 60–100 critical — are a declared reporting convention, not an empirically derived partition.

Released hours are converted into eliminable positions. The economic branch assumes that hours the AI removes translate into people the organization stops needing. In practice an organization may reassign those people, absorb growth without hiring, or reduce overtime instead of headcount. In all of those cases the saving exists but does not take the form the model gives it. P' should therefore be read as labour capacity no longer required, not as a redundancy instruction, and where reduction is not intended S is the ceiling of what could be saved rather than what will be.

The multiplier is applied to AI spend, not to outage impact. λ surcharges the cost of the AI because that is a figure the organization holds, whereas the cost of non-operation generally is not. This produces a counter-intuitive artefact across architectures of very different cost: a safer, expensive on-premises deployment can carry a larger absolute premium than a fragile, cheap API. Within one architecture the comparison is sound, which is how the model is used; across architectures the index rather than the monetary premium is the quantity to compare. Anchoring the premium to an hourly cost of non-operation is the most important pending refinement.

Scope. The model addresses cost and operating continuity. It does not model revenue lost during an outage, reputational damage, regulatory penalties, or the risk that a provider degrades a model without withdrawing it, which the June 2026 episode suggests deserves separate treatment. Two organizations with the same IDA can therefore face very different losses from the same interruption: the index measures exposure to an outage, not the cost of one.

The index is blind to margin above the floor. Two organizations with the same a_{ef} and the same λ receive the same index whether the retained workforce sits well above the continuity floor or exactly on it, and the index likewise does not react to the boundary case $P_{min} > P$ treated in Section 3.4, although the model reports that case separately. Distance to the floor is operationally meaningful and the present formulation does not carry it. The natural repair is a second reported quantity rather than a change to the index: a continuity margin $CM = (P' - P_{min})/P'$, which is zero when the retained workforce sits exactly on the floor and approaches one when it has room; it is undefined for $P' = 0$, the case of an area fully automated with no declared floor, and should be reported as not applicable there. That definition carries a blind spot worth declaring before anyone implements it: since P' equals P_{min} whenever the floor binds, CM returns zero both for an organization sitting exactly on the floor and for one already below it, which are opposite situations. Telling them apart requires comparing current headcount P against the floor, not the retained P'. Reported beside the index rather than folded into it, it would preserve the separation of saving from exposure that Section 3.7 defends, while restoring the information the index discards. It is not implemented here, and it is offered as the specific extension this limitation calls for.

The floor is required capacity, not available capacity. P_{min} is derived from the hours the organization declares it must cover, and the model assumes the retained people can still cover them. That assumption weakens precisely as automation proceeds: skills that are not exercised degrade, and an operator who has supervised a system for two years is not the operator who ran the process by hand before it. The model therefore gives a lower bound on the workforce required, and the gap between required and available capacity is outside its scope. Quantifying that gap is the natural continuation of this work.

The same quotient carries a second and stronger assumption: dividing hours by hours treats people as **interchangeable**. In practice H_{min} is rarely a homogeneous block. If 240 of the 640 hours in the case can only be covered by someone holding a specific qualification, the real floor is not ceil(640/160) = 4 but the sum of the per-skill floors, which may be larger. The correct reading is therefore to apply the floor per skill family and sum, and the single number the model reports holds when the critical hours are reasonably substitutable for one another. Where they are not, P_{min} underestimates.

The June 2026 episode was model-specific, not provider-specific. The directive affected two named models; the provider's other models remained available [6], so an organization holding a rehearsed fallback to a different model from the same provider continued to operate. Table 2 prices infrastructure concentration and carries no premium for dependence on a single model. The episode that motivates this paper is therefore not fully captured by the model the paper proposes: an organization running one model from one frontier provider is more exposed than its index indicates. A model-concentration premium is the first structural addition the table requires. The supervisory data point the same way: in the 2024 UK survey the top three model providers accounted for 44 per cent of all reported model providers against 73 per cent for cloud, and the model figure had risen from 18 per cent in 2022 while the cloud figure fell [9]. Concentration at the model layer is of the same order as at the infrastructure layer and is growing faster, whereas Table 2 prices only the latter. No provisional value is proposed for such a premium here. The evidence bounds its order of magnitude and not its size, and asserting a coefficient for the one dependency this paper has just shown it cannot yet measure would repeat, in a new place, the weakness Section 4.5 exists to bound.

The part of r a decision can undo. The model treats r as an input of the same kind as the others, and Section 3.2 shows that it is not: the policy term r_p is revocable and the regulatory term r_n is not. The index does not distinguish between them, so two organizations with the same IDA may retain human hours of very different durability, and the one leaning on policy is exposed to a risk this work does not quantify: that of its own future decision. The split is reported beside the index rather than inside it, for the same reason as the continuity margin — folding it in would require a probability of revocation that nobody can estimate today without inventing it. Acknowledging the limit and surfacing the input is all that can properly be claimed here.

External backup cannot be counted twice. R is declared for one area, and the model treats it as available to that floor. When a single organization evaluates several processes, the same people frequently appear as backup in more than one, and the resulting indices can no longer be read together: each process reports an exposure reduced by capacity it does not hold exclusively. The worst case is also the likeliest, because a provider outage strikes every process **at once**, which is exactly when shared backup does not stretch to cover both. So long as the model is applied to one process at a time the problem does not arise; on aggregation, R must be divided among the processes invoking it rather than replicated in each.

Terminology. The phrase "AI dependency" is also used in the human–computer interaction and psychology literatures for an individual user's reliance on an AI system. The present index measures something distinct: an organization's operational dependence on an external provider. The two are unrelated and should not be conflated. The offloading score of Padmakumar et al. [15], which estimates the fraction of a task's steps a user delegates to a tool by constructing a counterfactual workflow, is the individual-level measure. It is conceptually the task-level analogue of a_{ef} here, and where a process is observable it could serve as an instrument for estimating a and η rather than eliciting them.

7. Conclusion

Organizations that depended on those two models without a rehearsed fallback faced an abrupt interruption of nearly three weeks in June 2026, with no infrastructure failure, and the capacity they had left depended on the fallbacks they had retained, human capacity included. They discovered, at the worst possible moment, a number they had never computed. This paper proposes that the number be computed in advance, using inputs the organization already has, and reported alongside the saving rather than instead of it.

The model reframes the human factor from a declared organizational value into measurable continuity infrastructure, and gives decision-makers a single defensible figure to bring to a vendor negotiation. Its principal claim is modest and testable: that past a floor which the organization itself declares, additional automation stops buying savings and continues buying exposure, and that both behaviours follow from the equations rather than from narrative. Section 4.5 establishes that this claim, and the premium-induced ordering of configurations at fixed AI expense, survive a ±50 per cent perturbation of every risk coefficient in the model; verdicts on an individual project do not, and are reported as a threshold for that reason.

The model is offered as a proposed quantitative framework, not as a validated index. Its coefficients await calibration and its predictions await empirical cases. Both the implementation and the test battery are public precisely so that the refutation, if it comes, can be specific.

The architecture premiums of Table 2 and the backup discount are the first quantities that should change. Practitioners who run the model on their own operation, and organizations able to contribute anonymised incident data, are invited to write to the author: calibrating those values against observed cases is what would move the IDA from a proposed framework to a validated one.

The direction of regulation is compatible with the problem addressed here, though it is no evidence for the method proposed to address it. A parliamentary committee has already recommended designating the major AI providers as critical third parties [12], and a designated dependency is one that can be asked for a sized fallback rather than a documented intention.

6.1 Two consequences for whoever decides

The payback horizon is not neutral. T_{max} enters the model as a policy of the organization, and the model holds no view on its value. It is worth noting, however, what the subject of study does to that parameter: the technology being bought changes in price and capability over short periods, so a project that needs eighteen or twenty-four months to pay for itself is computing the payback of a configuration that will probably no longer be current by the time it is paid off. The reference case is exactly that: the project recovers in twenty months and therefore does not pass. This is not a rule of the model, but it is a defensible reading of its results: **the faster the contracted layer moves, the shorter the T_{max} it should be judged with.**

When bringing the AI in-house pays. The same arithmetic answers a question boards ask often. Running the models on own infrastructure carries a fixed cost and a low multiplier; using an API carries a variable cost and a high one. Own infrastructure pays when $\lambda_{own} \cdot C_{own} < \lambda_{api} \cdot c \cdot H \cdot a_{ef}$, that is, when the automated volume is large enough for the variable cost to exceed the fixed one.

Rearranged, the threshold is $H \cdot a_{ef} = \lambda_{own} \cdot C_{own} / (\lambda_{api} \cdot c)$. What decides it is therefore not the hardware but **the price of the API hour**, since the threshold is inversely proportional to c. With the hardware of the case — 8,420 a month, $\lambda = 1.02$ against multi-cloud with a plan at 1.10 — and $c = 2$, the threshold sits at **3,904 AI hours per month**: some 24 people working on it full time, or about 64 in the area under the reference configuration, where only 38.4 per cent of hours is automated. With a low-priced model, of the order of 0.20 an hour, the same hardware needs ten times the volume to justify itself.

The practical reading is the same at both ends. With the ten people of the case the API imputes 1,352 a month against 8,588 for own infrastructure, and the gap is wide enough that no choice of premiums reverses it. **For the great majority of areas, bringing the AI in-house is not justified on cost, and the right answer to the risk is not to move but to spread: multi-cloud with a rehearsed fallback.** Own infrastructure begins to pay only at scale, or where the data cannot leave the organization — which is a legal reason and not an economic one, and the model does not arbitrate it.

Use of artificial intelligence tools

The author declares having used artificial intelligence assistants — Claude, by Anthropic — throughout this work: to discuss the formulation of the model, review its internal consistency, implement and verify the reference calculator and the edge-case battery, and draft and correct parts of the manuscript. The modelling decisions, the choice of coefficients, the interpretation of results and the final wording are the author's, who is answerable for the content in full. This statement is included because several publication venues require it, and because the subject of this work makes omitting it incoherent.

The tools used are not listed as authors. Authorship entails answering for the work to those who read, criticise or cite it, and an AI system cannot carry that responsibility.

Availability

Interactive model: https://modelo-ida.web.app · **English edition:** https://modelo-ida.web.app/en
Source code and test battery: https://github.com/All-Aideas/Modelo-IDA
Archived: https://doi.org/10.5281/zenodo.22070415 — concept DOI, always resolves to the latest deposited version
Licence: MIT. No registration, no server, no data leaves the browser.

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